Teacher Information, Course Description and Lecture-wise Plan

of

Artificial Intelligence (CSC262)

under

Pilot Project:

Uniform Teaching, Assessment and Quality Assurance Across All The Campuses

By

Faculty of Information Science and Technology
COMSATS University Islamabad

A: Teacher Information

Course Name: Artificial Intelligence	Credit Hours: 3(2,1)
Course Code: CSC262	Pre-requisite: None
Teacher Name:	Teacher E-mail:
Program: BS (Computer Science)	Semester: Fall 2026
Department: Computer Science	Campus:

B: Course Contents

This course provides an in-depth treatment of the foundational, symbolic techniques of Artificial Intelligence, together with a broad overview of its major learning-based domains. Students will develop the ability to apply core AI techniques — intelligent agents, problem-solving and search (uninformed, informed, local, and adversarial, including evolutionary algorithms), constraint satisfaction, automated planning, knowledge representation, rule-based expert system architecture, logical inference, and reasoning under uncertainty — to solve AI problems. The course then surveys, at an introductory level, the learning-based domains of AI, including machine learning and deep learning, and natural language processing, to build student understanding of how these domains extend AI to real-world applications. The course concludes with a discussion of AI ethics and emerging trends.

Recommended Books:

1. Russell, S., and Norvig, P. (2020). Artificial Intelligence: A Modern Approach (4th ed.). Pearson.
2. Poole, D. L., and Mackworth, A. K. (2023). Artificial Intelligence: Foundations of Computational Agents (3rd ed.). Cambridge University Press. Freely available at https://artint.info

Unit-wise Major Topics (CDF V7.1)

Unit	Topic	Hours
1	Artificial Intelligence Foundation: Concepts, History; and Intelligent Agents & their Environments.	4
2	Problem-Solving: Problem-Solving Agents; Uninformed & Informed Search Strategies; Local Search (including Evolutionary Algorithms); Adversarial Search; Monte Carlo Simulation; and Constraint Satisfaction Problems (CSPs)	10
3	Knowledge Representation & Reasoning: KR using First-Order Logic; Inference in FOL; Rule-Based Expert Systems; and Uncertainty & Probabilistic Reasoning	6
4	Automated Planning: Classical Planning; State-Space & Partial-Order Planning; Planning Graphs; Hierarchical Planning.	4
5	Overview of Modern AI & Deep Learning: Learning Paradigms (Supervised, Unsupervised, Reinforcement Learning); and an Overview of Artificial Neural Networks (ANN) & DL.	4
6	Applications of AI	2
7	Emerging trends of AI, AI Ethics, Human-AI Interaction	2
	Total Contact Hours	**30**

Note on calendar: this plan schedules 32 numbered lecture slots across 16 teaching weeks (2 lectures/week), which is more than the 30 CDF-approved content hours. The two extra slots are the Mid Term Exam (Lecture 13, after Unit 2) and a dedicated Course Synthesis & Revision session (Lecture 32, before the Final). Neither consumes CDF-approved unit hours; both exist to absorb routine calendar disruption (Sports Week, holidays) without cutting into taught content, and to give a real, scheduled slot for revision.

C: Lecture-wise Plan

Week #	Lecture #	Topic (Chapter/Topics Covered)	Practice / Evaluation	Remarks
1	1 (Date)	***What Is AI? Foundations*** • *Four views of AI (acting/thinking humanly/rationally) – brief* • *Intellectual foundations (philosophy, maths, neuroscience, etc.) – overview only* • *Brief history: founding period → early enthusiasm → AI winters → statistical return*	*Discussion: everyday AI examples*	*AIMA Ch.1 §1.1–1.2*
	2 (Date)	***History, State of the Art, Risks & Benefits*** • *Deep-learning era (high-level)* • *What modern AI can / cannot do (state of the art)* • *Risks & benefits (safety, bias, employment) – introductory only; full ethics in Unit 7*	*Short timeline quiz / questions*	*AIMA Ch.1 §1.3–1.5*
2	3 (Date)	***Intelligent Agents: PEAS & Rationality*** • *Agents & Environments; PEAS description* • *Rationality vs omniscience* • *PEAS examples (vacuum world, taxi) – 1–2 only*	*Exercise: write PEAS for 1–2 agents*	*AIMA Ch.2 §2.1–2.2*
	4 (Date)	***Nature of Environments; Structure of Agents*** • *Key environment properties (observable, deterministic, episodic, static, discrete, known) – definitions + examples* • *Agent types: simple-reflex → model-based → goal-based → utility-based → learning agents* • *NOTE: This taxonomy is the unifying frame for the whole course*	***Quiz 1 (CLO-1)***	*AIMA Ch.2 §2.3–2.4*
3	5 (Date)	***Problem-Solving Agents; Uninformed Search*** • *Formulating problems: initial state, actions, transition, goal test, path cost* • *Core uninformed: BFS* • *Completeness, optimality, time/space – comparison table only*	*Trace BFS on small graph*	*AIMA Ch.3 §3.1-3.2*
	6 (Date)	***Problem-Solving Agents; Uninformed Search*** • *DFS, UCS, Iterative Deepening* • *Completeness, optimality, time/space – comparison table only*	*DFS/UCS on a small graph*	*AIMA Ch.3 §3.2–3.4*

4	**7** (Date)	***Informed Search & Heuristics*** • *Greedy Best-First; A* Search* • *Admissible & consistent heuristics (why they matter)* • *Effect of heuristic accuracy (intuition only)*	*Assignment 1 (CLO-1): A* with admissible heuristic*	*AIMA Ch.3 §3.5–3.6*
	8 (Date)	***Local Search & Optimization*** • *Hill-Climbing (steepest-ascent + random-restart)* • *Simulated Annealing (core idea + cooling)* • *Local Beam Search – brief*	*Practice: hill-climbing on simple landscape*	*AIMA Ch.4 §4.1–4.2*
5	**9** (Date)	***Evolutionary Algorithms – Part 1 (SPLIT)*** • *Evolutionary Algorithms as population-based local search* • *Genetic Algorithms: chromosome encoding, fitness function* • *Selection methods (roulette-wheel, tournament) – core idea*	*Lab-linked: simple GA setup*	*AIMA Ch.4 §4.1 (p.115–119)*
	10 (Date)	***Evolutionary Algorithms – Part 2 (SPLIT)*** • *Crossover & mutation operators* • *Convergence & premature convergence* • *Comparison with hill-climbing / simulated annealing; typical applications*	*Continue / finish GA exercise*	*AIMA Ch.4 §4.1*
6	**11** (Date)	***Adversarial Search: Minimax*** • *Two-player zero-sum games; game trees* • *Minimax algorithm (core)*	*Hand-trace minimax on a small tree*	*AIMA Ch.5 §5.1-5.2*
	12 (Date)	***Adversarial Search: Alpha beta pruning*** • *Alpha-Beta pruning (15–20 min focus – pruning idea + simple tree)*	*+ α-β on small tree*	*AIMA Ch.5 5.3*
7	**13** (Date)	***MCTS & Stochastic / Partial Observability (Overview)*** • *Monte Carlo Tree Search: Selection → Expansion → Simulation → Back-propagation (high-level)* • *Brief note on stochastic & partially observable games*	*Simulate 2–3 MCTS rollouts by hand*	*AIMA Ch.5 §5.4–5.5*
	14 (Date)	***Constraint Satisfaction Problems*** • *CSP definition: variables, domains, constraints; map-colouring / scheduling examples* • *Backtracking search + variable/value ordering heuristics (core)*	*Model one CSP; backtracking sketch*	*AIMA Ch.6 §6.1–6.5 (core only)*

		• *Min-Conflicts local search for CSPs*		
8	**15** (Date)	***Knowledge: Definition, Types & Knowledge-Based Agents*** • *What is knowledge? Types (declarative, procedural, meta, etc.) – brief* • *Knowledge-Based Agents; Wumpus World as running example* • *Propositional Logic: syntax, semantics, entailment, simple KB (core only)*	*Build a tiny propositional KB*	*AIMA Ch.7 §7.1–7.2, 7.4*
	16 (Date)	***Propositional Theorem Proving*** • *Inference rules; conversion to CNF* • *Resolution (core algorithm + example)* • *Brief note on DPLL / SAT (existence only)*	*Convert to CNF + resolve*	*AIMA Ch.7 §7.5–7.6*
9	**17** (Date)	***MID-TERM EXAMINATION***	***Midterm (25%)***	*After ≈ 8 weeks*
	18 (Date)	***First-Order Logic: Syntax, Semantics, KE*** • *Objects, relations, functions, quantifiers* • *Using FOL to represent domain facts* • *Knowledge-engineering process (high-level steps)*	*Translate English → FOL sentences*	*AIMA Ch.8 §8.1–8.4*
10	**19** (Date)	***Rule-Based Expert Systems + Forward/Backward Chaining*** • *Architecture: KB, Working Memory, Inference Engine, Explanation, UI* • *Production systems: forward-chaining (data-driven) vs backward-chaining (goal-driven)* • *Conflict resolution; brief case (MYCIN / CLIPS)*	*Assignment 2 (CLO-2): small rule-base design*	*Poole & Mackworth Ch.14; AIMA inference sections*
	20 (Date)	***Inference in FOL (Linked to Chaining)*** • *Unification & lifting (core idea)* • *Forward & Backward chaining in FOL context* • *Resolution in FOL (high-level)*	***Quiz 2 (CLO-2)***	*AIMA Ch.9 §9.1–9.5 (core)*
11	**21** (Date)	***Uncertainty & Probabilistic Reasoning*** • *Acting under uncertainty; basic probability notation* • *Bayes' Rule; Naïve Bayes (core diagnostic example)* • *Bayesian Networks – semantics only (introductory)*	*Apply Bayes' Rule to a diagnostic scenario*	*AIMA Ch.12–13 (selected)*

	22 (Date)	***Classical Planning: Definition & Representation*** *• Planning problem; STRIPS / PDDL-style representation (core)* *• Example domains: blocks world / air cargo (one worked)*	*Formalise a simple domain in STRIPS*	*AIMA Ch.11 §11.1*
12	**23** (Date)	***State-Space Planning & Planning Graphs*** *• Forward (progression) & Backward (regression) search* *• Planning Graphs for heuristic estimation (core construction)*	***Quiz 3 (CLO-3)***	*AIMA Ch.11 §11.2–11.3*
	24 (Date)	***Hierarchical Planning & Nondeterministic Overview*** *• Hierarchical Task Networks: high-level actions → refinement* *• Brief note on sensorless / conformant planning & resources*	*Assignment 3 (CLO-3): hierarchical task decomposition*	*AIMA Ch.11 §11.4–11.5*
13	**25** (Date)	***Artificial Neural Networks deep learning Overview*** *• Introduction of Artificial Neural Networks and deep learning*		*AIMA Ch.21 §21.1*
	26 (Date)	***Artificial Neural Networks deep learning Overview*** *• Applications of Artificial Neural Networks and deep learning*	***Quiz 4 (CLO-4)***	
14	**27** (Date)	***AI Applications*** *• Use Case of AI in Text and Computer Vision*		
	28 (Date)	***AI Applications*** *•Use Case of AI in Drug discovery, Speech recognition and generation, robotics, transportation. etc.*	*Assignment 4 (CLO-4)*	
15	**29** (Date)	***Emerging trends of AI***		
	30 (Date)	***AI Ethics, Human-AI Interaction***		
16	**31** (Date)	***Course Synthesis & Revision*** *• Revisit agent framework across all CLOs*	*Open Q&A / practice problems*	*Outside CDF 30-hour content budget*

		• *Key algorithms & worked problems for Final Term*		
	32 (Date)	*Review*		
17	(Date)	***FINAL TERM EXAMINATION***	***50%***	

Note: All Assignments/Quizes weightage is 25%